Opportunity Win Probability Prioritisation Model

A Data-Driven Pre-Sales Triage Framework

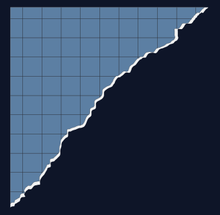
Ploypalin Tan

Enabling structured capacity allocation under constrained solution bandwidth

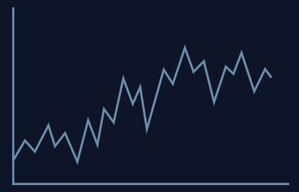
Pre-Sales Capacity Is Finite, but Opportunity Volume Is Not



Pre-sales bandwidth is structurally constrained



Opportunity inflow continues to scale



Manual prioritisation ***lacks*** consistency



High-potential deals risk *under-allocation*

The problem is not volume

The problem is disciplined triage

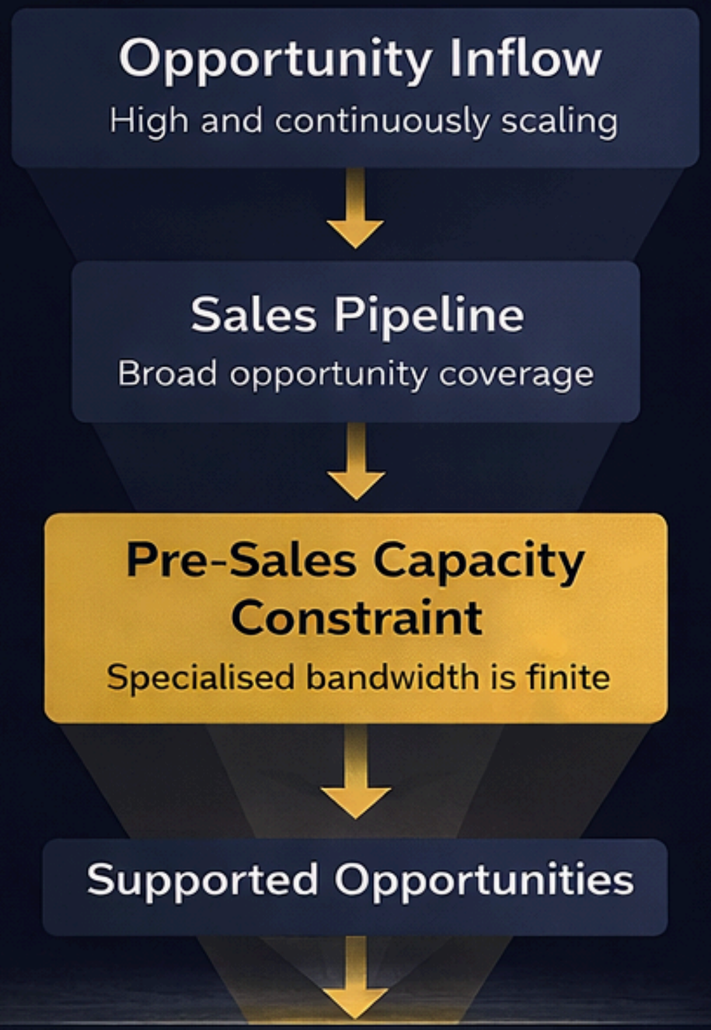
This Framework Is Designed for Capacity-Constrained Decision Makers

1 PRIMARY USER

Solutions Consultant / Pre-Sales

Operational Constraint

- Finite specialised solution bandwidth
- Deep technical support cannot scale with inflow
- Manual triage creates inconsistent coverage
- High-potential deals risk under-allocation



2 SECONDARY STAKEHOLDER

Sales Manager / RevOps

Commercial & Operating Risk

- Limited signal on true opportunity quality
- Resource allocation lacks systematic prioritisation
- Pipeline breadth may dilute win concentration
- Forecast confidence impacted by uneven deal support

Structural tension between demand and specialised capacity

The model augments human judgement – it does not replace it.

Capacity Constraints Demand Concentrated Win Density

CONSTRAINTS

- Only attributes available at opportunity creation
- No behavioural execution signals
- Pre-sales bandwidth is finite

DESIGN PRINCIPLES

- • Ranking over prediction perfection
- • Optimise top-decile concentration

SUCCESS METRIC

Precision @10%

Increase win concentration within the supported segment

Workload Concentration Occurs Where Signal Quality Is Weakest

CONVERSION QUALITY

Larger accounts demonstrate modestly higher **win rates**.



Small Accounts (<10k)	Mid Accounts (10-50k)
57.4%	58.7%

WORKLOAD DISTRIBUTION

Capacity pressure is structurally dominated by **small-account volume.**



Small Accounts (<10k)	Mid Accounts
7,179	196

Structural workload imbalance - *not* conversion spread - creates the prioritisation **challenge**.

Economic Fit, Not Absolute Deal Size, Drives Early Conversion Signal

Raw View

Deal Size:

$40,000

Company Revenue:

$500,000

Deal Size:

$50,000,000

Same-looking deal category, completely different economic burden.

Scaled-Normalised View

Deal/Revenue Ratio

8% vs **0.08%**



Big commitment
harder to convert

Tiny commitment
easier to convert

Absolute deal size ignores company capacity.

Why Precision at the Top Matters

GLOBAL EVALUATION METRICS

AUC

Evaluates how well the model separates wins from losses

Accuracy

Captures overall prediction correctness, regardless of prioritisation needs.

TRIAGE-ALIGNED METRIC

Precision@10%

Evaluates conversion concentration within the deployable capacity segment.

Lift@10%

Measures the incremental prioritisation value over random allocation.

In a capacity-constrained environment, performance at the *top* of the ranking curve matters most.

Top-Decile Precision Demonstrates Measurable Triage Lift

BASELINE MODEL

Precision@10%

0.551

Lift@10%

0.872

AUC: 0.487

Limited concentration of wins within top decile.

FEATURE-ENGINEERED MODEL

Precision@10%

0.672 ★

Lift@10%

1.079 ★

AUC: 0.502

Material improvement in win density at deployable capacity.

ALTERNATIVE MODEL

Precision@10%

0.582

Lift@10%

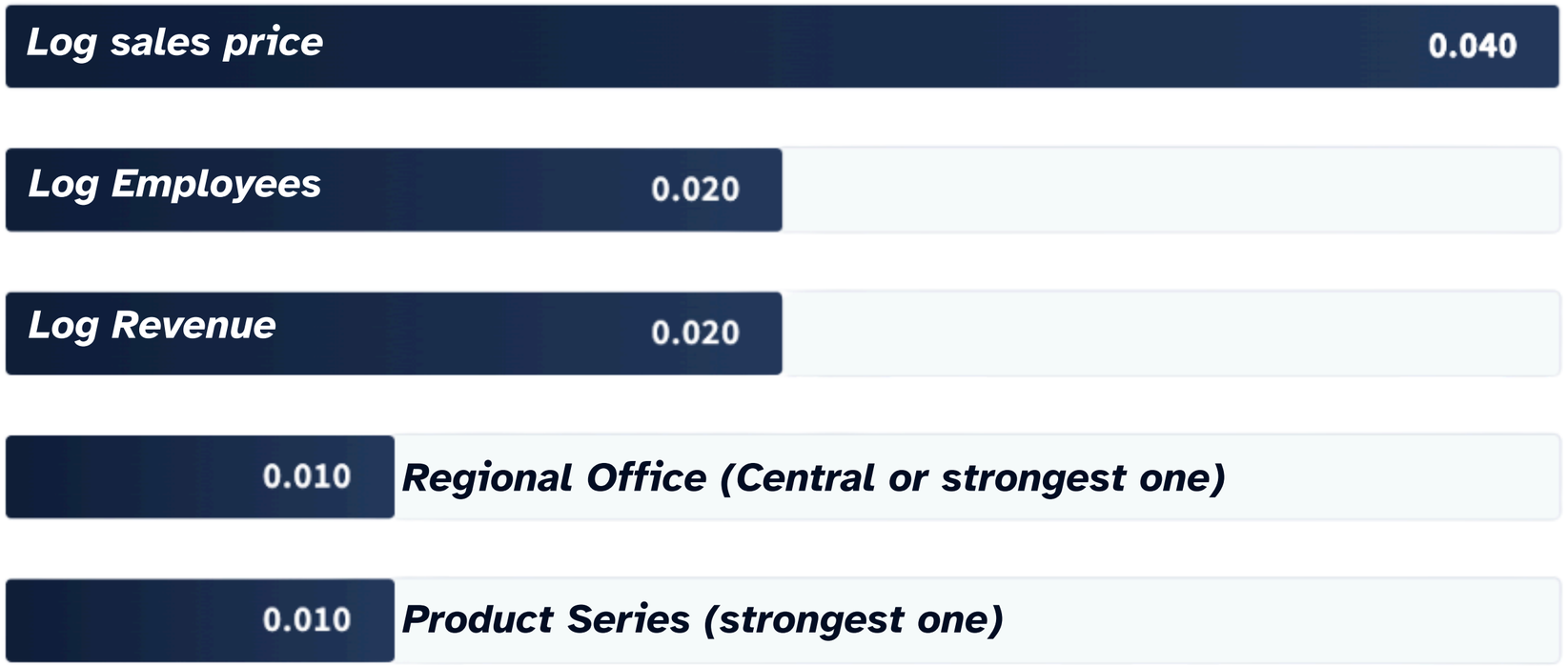
0.935

AUC: 0.505

Marginal improvement, but weaker top-decile alignment.

Optimisation for top-decile precision yields materially higher triage value than baseline and alternative models.

Firmographic Scale Dominates Early-Stage Predictive Signal



Mean Absolute Impact on Win Probability
aka percentage points impact on win probabilit.

Log transformation applied to normalise scale and reduce skew.

- **Log sales price** is the strongest global driver of conversion probability
- Firmographic scale (revenue and employees) materially influence outcomes
- Regional and product effects are *secondary contributors*
- Economic capacity signals outweigh sector and regional effects

Model Architecture Influences Relative Feature Emphasis

Deployment Model

Logistic Regression



Linear weighting

Supporting Robustness Model

Random Forest



Interaction learning

Sector and regional dummies receive strong directional coefficients

Firm size and economic alignment features rank highest in importance

Economic capacity remains structurally influential across model families.

Model Selection Rationale

Deployment Model

Logistic Regression



Linear weighting

interpretability &
deployment stability

Supporting Robustness Model

Random Forest



Interaction learning

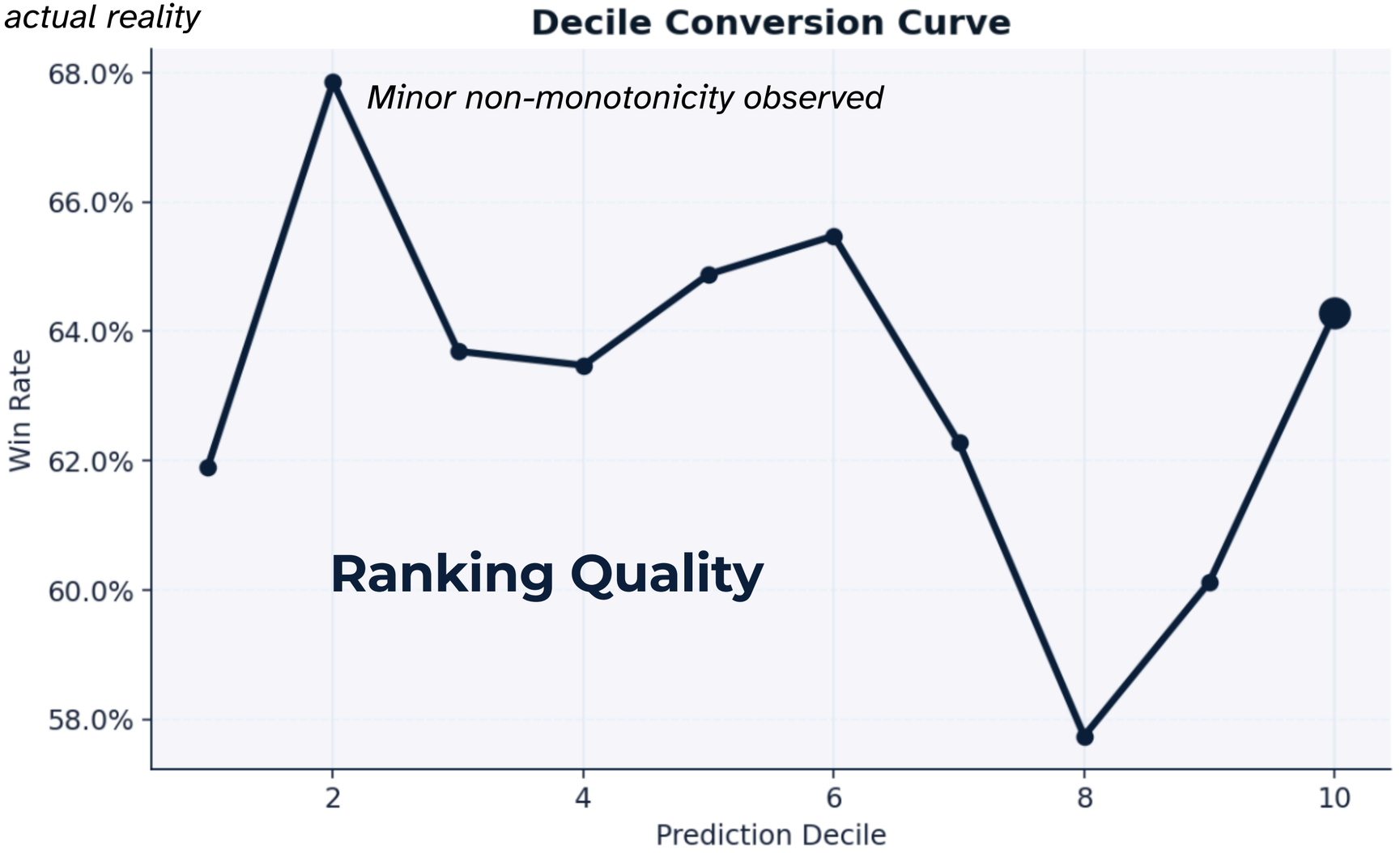
validates structural
signal robustness

Both confirm economic capacity dominance

Model Demonstrates Moderate Ranking Ability

Moderate separation is sufficient for prioritisation

actual reality



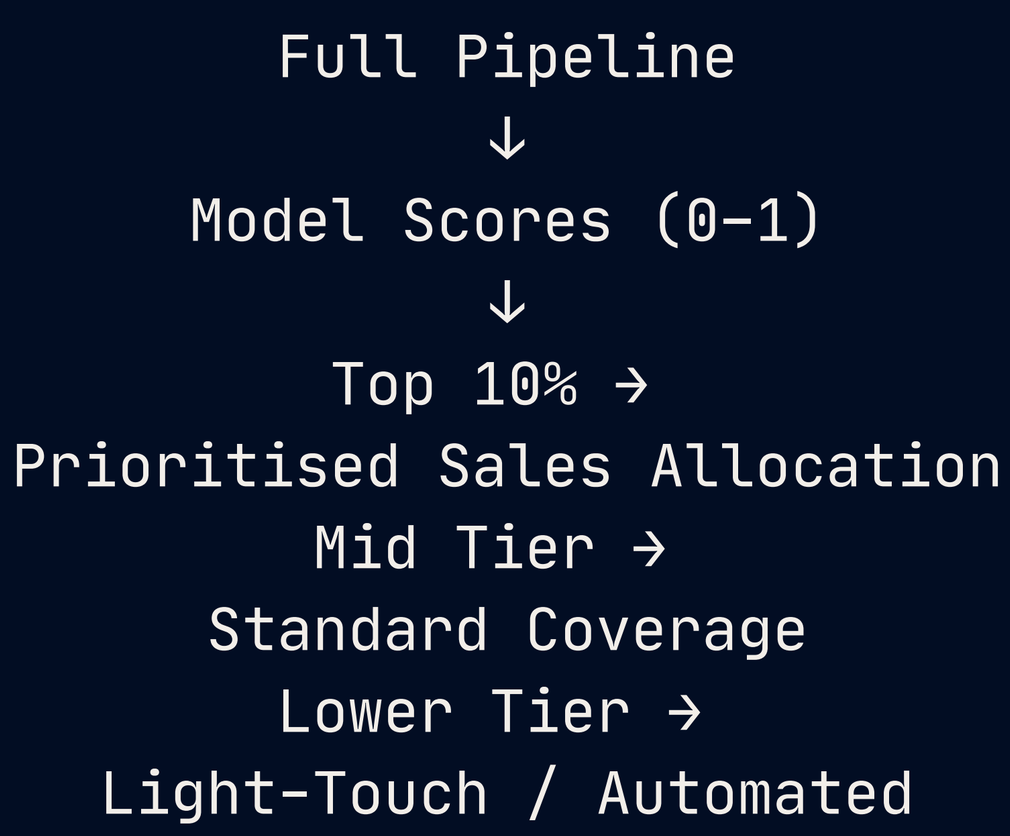
checks whether model correctly ranks who is more likely to convert.

The decile curve evaluates the model's ability to rank opportunities by likelihood of conversion. While some uplift is observed, the separation remains modest.

Controlled Deployment Architecture

Win probability informs prioritisation not automation.

Operating Logic (Visual)



Guardrails (Governance Boundaries)

- No automatic approval / rejection
- Human override preserved
- Behavioural execution signals not captured

★ ***The model enforces capacity discipline while preserving commercial judgment.***

Precision@10% Monitoring Framework

Conceptual illustration of how Precision@10% should be tracked post-deployment



Performance discipline must match deployment discipline.

Model Constraints & Governance Considerations

Structural Constraints

- Early-stage dataset lacks behavioural execution signals
- Moderate AUC indicates directional, not deterministic, signal
- Model performance sensitive to pipeline mix

Mitigation Controls

- Precision@10% threshold governance
- Drift monitoring & recalibration cadence
- Human override retained

The model informs prioritisation; it does not replace commercial judgement.

Phased Deployment Roadmap

1

Phase 1 — Controlled Rollout

- **Shadow scoring**
- **Top-decile triage pilot**
- Precision baseline established

2

Phase 2 — Governance Layer

- Monthly monitoring cadence
- Threshold optimization
- Sales feedback loop

3

Phase 3 — Signal Expansion

- Behavioural feature enrichment
- Execution velocity tracking
- Model recalibration

Structured prioritisation today. **Signal enrichment tomorrow.**

Operational Impact by Persona

1 PRIMARY USER

Solutions Consultant / Pre-Sales

Precision@10% improvement

Baseline: 0.551
Selected model: 0.672

AFTER:

Precision@10: 55% → 67%

Pre-sales efficiency



2 SECONDARY STAKEHOLDER

Sales Manager / RevOps

Lift@10% improvement

Baseline: 0.872
Selected model: 1.079

AFTER:

Lift@10: 0.87 → 1.08

pipeline quality concentration

So What Does This Actually Mean?

I

We prioritise where capacity converts.

II

The model supports judgment - it does not automate decisions..

III

We monitor and recalibrate to protect lift.